

JEE PHYSICS FORMULAS & DERIVATIONS

Comprehensive Quick-Reference Sheet (Kinematics, Rotational Dynamics, & Waves)

1. Kinematics & Projectile Motion

Equations of Motion (Constant Acceleration): These linear differential derivations relate displacement (s), initial velocity (u), final velocity (v), acceleration (a), and time (t) under constant gravity field assumptions.

Velocity-Time Relation	$v = u + a * t$
Displacement-Time Relation	$s = u*t + 0.5 * a * t^2$
Velocity-Displacement Relation	$v^2 = u^2 + 2 * a * s$

Projectile Motion: In a uniform two-dimensional coordinate framework with horizontal launch angle θ , and constant gravity vector g .

Time of Flight (T)	$T = (2 * u * \sin(\theta)) / g$
Maximum Height (H)	$H = (u^2 * (\sin(\theta))^2) / (2 * g)$
Horizontal Range (R)	$R = (u^2 * \sin(2 * \theta)) / g$
Equation of Trajectory	$y = x * \tan(\theta) - (g * x^2) / (2 * u^2 * (\cos(\theta))^2)$

2. Rotational Dynamics & Moment of Inertia

Moment of Inertia (I): Represents the rotational inertia about a specified rigid body rotational axis, calculated via integral $r^2 dm$.

Thin Ring (Radius R)	$I = M * R^2$
Solid Disc (Radius R)	$I = 0.5 * M * R^2$
Solid Sphere (Radius R)	$I = 0.4 * M * R^2$ [2/5 MR ²]
Hollow Sphere (Radius R)	$I = 0.67 * M * R^2$ [2/3 MR ²]
Thin Rod (Length L, Center)	$I = (1/12) * M * L^2$

Theorems & Kinetic Equations: Transforming translational constraints into angular system parameters.

Parallel Axis Theorem	$I = I_{cm} + M * d^2$
Perpendicular Axis Theorem	$I_z = I_x + I_y$ [For planar laminas]
Torque Equation	$\tau = I * \alpha = r \times F$
Angular Momentum	$L = I * \omega = r \times p$
Rolling Kinetic Energy	$K = 0.5 * M * v^2 + 0.5 * I * \omega^2$

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Part 2: Oscillations, Waves & Acoustic Doppler Principles

3. Simple Harmonic Motion (SHM)

Simple Harmonic Motion: Governed by a linear restoring force proportional to displacement. The motion exhibits a sinusoidal pattern centered around a stable equilibrium state.

Differential Equation	$d^2x/dt^2 + \omega^2 * x = 0$
Angular Frequency (ω)	$\omega = \sqrt{k / m} = 2 * \pi * f$
Displacement (x)	$x(t) = A * \sin(\omega * t + \phi)$
Velocity (v)	$v(t) = A * \omega * \cos(\omega * t + \phi) = \pm \omega * \sqrt{A^2 - x^2}$
Acceleration (a)	$a(t) = - (\omega^2) * x(t)$
Simple Pendulum Period	$T = 2 * \pi * \sqrt{L / g}$

4. Wave Mechanics & Doppler Effect

Wave Equations: Mathematical representation of localized energy propagation through spatial-temporal coordinates.

Wave Equation	$y(x, t) = A * \sin(k * x - \omega * t + \phi)$
Wave Number (k)	$k = 2 * \pi / \lambda$
Wave Speed (v)	$v = f * \lambda = \omega / k$

Doppler Effect: The apparent shift in acoustic pitch frequency detected by an observer moving relative to a sound emitter wave source.

General Doppler Formula	$f' = f * [(v \pm v_{\text{observer}}) / (v \mp v_{\text{source}})]$
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Rule of Signs: Use the upper signs when the source/observer move toward each other (frequency rises). Use the lower signs when they move away (frequency drops).